PALEOMAGNETIC EVIDENCE ON THE GENESIS OF THE HARD HEMATITE ORE DEPOSITS OF THE VERMILION RANGE, MINNESOTA

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Paleomagnetic measurements using standard methods were made on 55 samples of hard hematite iron ore and associated host rocks that were collected from along the surface exposure, and from underground in the Pioneer Mine at Ely and in the Soudan Mine at Soudan, on the Vermilion Range, Minnesota, U.S.A. The unaltered host rocks have a stable pre-folding remanence of probable Keewatin age, whereas the ore and adjacent altered host rocks have a stable post-folding remanence of possibly Early to Middle Keweenawan age. This result indicates that the ore originated as a post-folding epigenetic deposit.

INTRODUCTION

Paleomagnetism is useful in studying the genesis of hard hematite ore deposits. By comparing the remanence directions of the ore and its host rocks, and by determining if the remanences were acquired before or after folding, it is often possible to determine the order and timing of the events of ore genesis. Gross and Strangway (1961), R. L. DuBois (1961, 1962) and Symons (1967) have used this method to study the genesis of hard hematite ore deposits.

The genesis of the hard hematite ore deposits at Ely and Soudan on the Vermilion Range, Minnesota, U.S.A. (Fig. 1) has been disputed. Winchell and Winchell (1891) proposed that they are anamorphosed primary chemical sediments. Clements (1903) and Abbott (1907) proposed that iron-bearing groundwater solutions metasomatically replaced the primary iron-formation in structural troughs. Van Hise and Leith (1911) argued that groundwater leached and oxidized the primary iron-formation to form residual soft hematite deposits, which at Soudan were metamorphosed directly to hard hematite and at Ely were weathered to laterites and then metamorphosed. Royce (1942) also argued that meteoric solutions created residual soft hematite deposits, which were subsequently metamorphosed to hard hematite. Gruner (1926, 1937) believed that hydrothermal solutions from intrusive rocks or groundwater heated by intrusive rocks oxidized, leached, and metamorphosed, brecciated magnetite iron-formation to form hard hematite ore. Roberts and Bartley (1943) suggested that ascending hydrothermal solutions oxidized, leached, and supplied additional iron from an igneous source by metasomatic replacement of the iron-formation to form hard hematite ore. Schwartz and Reid (1955), Reid 1956*a*, 1956*b*), Klinger (1956), and Machamer (1964) favoured post-deformational replacement of the iron-bearing members in the Soudan iron-formation by iron carried by ascending hydrothermal solutions.

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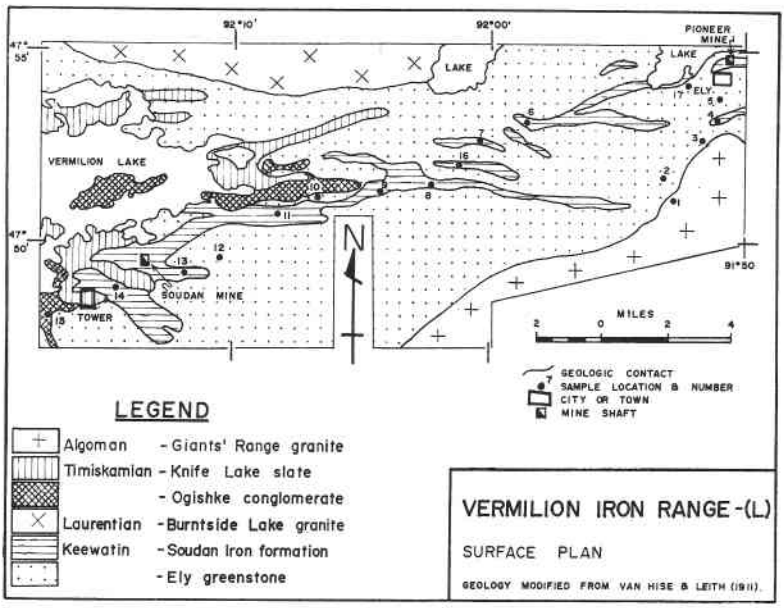


FIG. 1. Geology and surface sampling of the Vermilion Range.

A summary of the paleomagnetic methods used in this study is given in Symons (1967) and a more detailed account in Symons (1965). The prefix letters L, M, and N before sample numbers indicate that the samples came from the surface of the Vermilion Range, the Pioneer Mine, Ely, and the Soudan Mine, Soudan, respectively. The surface samples came from road cuts and the rest from underground.

GEOLOGY AND PALEOMAGNETIC RESULTS

General

Winchell and Winchell (1891), Clements (1903), Abbott (1907), Van Hise and Leith (1911), Gruner (1926), and Reid (1956*a*) have reported on the geology of the Vermilion Range, Reid (1956*b*) and Machamer (1964) on the Ely deposit, and Schwartz and Reid (1955) and Klinger (1956) on the Soudan deposits. The stratigraphy of the Vermilion Range shown in Table I is taken from Reid (1956*a*) and Goldich *et al.* (1961).

The regional geologic structure is not well understood because the lack of stratigraphic marker horizons has made correlations difficult. The Ely greenstones are exposed along a belt 6 to 12 miles wide and 40 miles long. The Soudan iron-formation, the Ogishke conglomerate, and the Knife Lake slates are found as infolded sediments within the Ely greenstones along the central axis of its belt of exposure, and form a synclinorium with axial planes striking N 70° E (Fig. 1). Within the synclinorium, there are numerous tight folds and faults. At Ely (Fig. 2), the hard hematite ore is found as a folded tabular body around the basal part of a tight 'U'- or 'V'-shaped synclinal fold, which forms a

TABLE I

Stratigraphy of the Vermilion Range

Era	Age (b.y.)*	Period – System (orogeny)	Vermilion Range
Mesozoic	.1	Cretaceous	Clastic sediments
	1.0	(Grenville)	unconformity
Late Precambrian		Keweenawan	Duluth gabbro, intrusive rocks, and sediments
	1.7	(Penokean)	unconformity
Middle Precambrian		Animikie Upper	Rove (Virginia) slates
			unconformity
		Middle	Gunflint (Biwabik) iron-formation
			unconformity
Early Precambrian	2.5	(Algoman)	Vermilion (Giant's Range) granite
			intrusive contacts
		Temiskaming	Knife Lake slates with basal Ogishke conglomerate member
			unconformity
		Keewatin	Soudan iron-formation
			Ely greenstone

*b.y.—billion (10^9) years.

trough $1\frac{3}{4}$ miles long, $\frac{1}{4}$ mile wide, and at least 1 600 ft deep. The samples were collected across the fold. There are several hard hematite ore deposits at Soudan. The ore deposits are folded tabular bodies up to 100 ft thick, up to 1 500 ft in horizontal length, and up to 2 500 ft in vertical length. One such deposit, the Montana, was sampled around a tight, westerly-plunging 'S' fold (Fig. 3).

The magnetic measurements made on the samples are summarized in Table II.

Ely Greenstone

The Ely greenstone contains several thousand feet of massive basaltic flows with minor fragmental and tuffaceous zones. The flows have been foliated and regionally metamorphosed to the chlorite facies.

Six samples of Ely greenstone (L–2, 3, 5, 10, 12, and 17) were collected on the surface (Fig. 1). These samples have fairly uniform magnetic characteristics and moderately stable remanences of weak intensity (Table II), which may indicate a thermal remanent magnetization (t.r.m.) or chemical remanent magnetization (c.r.m.). The precision parameter for the samples is low (3.5) prior to unfolding, and the value after unfolding was not determined because reliable tectonic corrections could not be made. There is no apparent relationship between the remanence directions and the foliation planes of the samples.

Three samples of Ely greenstone (M–1, 12, and 21) were collected from within 100 ft of the ore in the Pioneer Mine (Fig. 2). They have stable remanences of very weak intensity after a-c. demagnetization (Fig. 5, sample M–12) and very low ratios of remanent to unit-field induced intensity (Table II), which probably explains in part the low values for the precision parameter. Unfolding causes the precision parameter for the remanence directions to

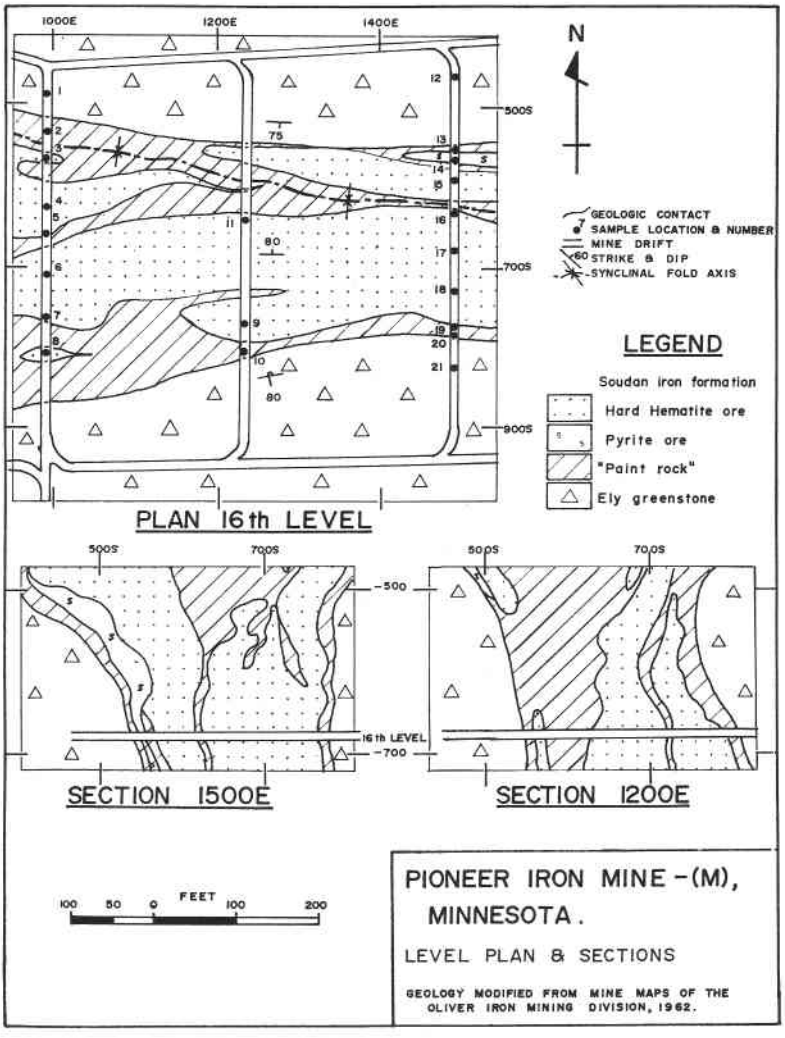


FIG. 2. Geology and sampling of the Pioneer Mine.

decrease in value from 2.0 to 1.8 (Fig. 4*b*). Both values are too low to be statistically significant and one explanation for them could be that the remanence had a mixed origin.

Soudan Iron-Formation

The Soudan iron-formation contains several stratigraphic horizons of iron-formation interbedded with andesitic and tuffaceous flows, and has an aggregate thickness of several hundreds of feet. The iron-formation is composed of magnetite-banded and hematite-banded oxide facies (James 1954), with minor amounts of iron silicates, iron carbonates, and pyrite. Jasper (i.e. jaspilite or banded red hematitic chert and blue specular hematite) is common in the iron-formation near the ore deposits.

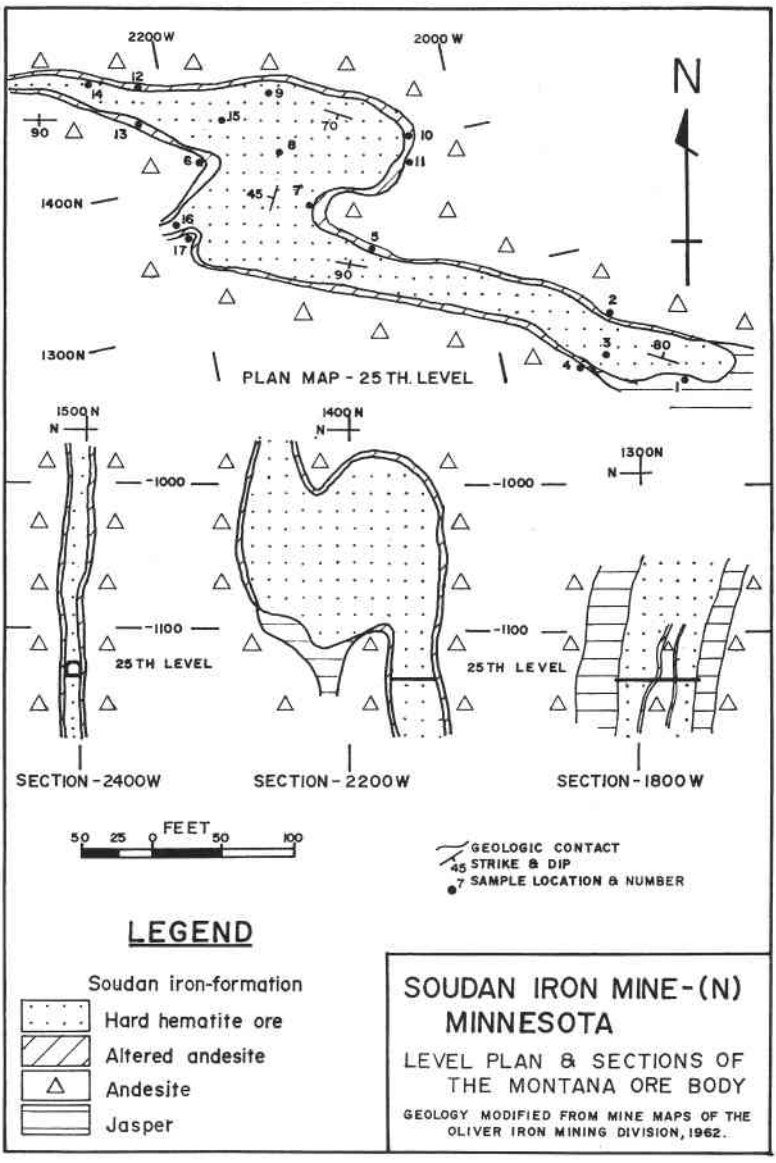


FIG. 3. Geology and sampling of the Soudan Mine.

Nine samples (L–4, 6 to 9, 11, 13, 14, and 16) were collected on the surface (Fig. 1) from the Soudan iron-formation, of which one andesite sample (L–4) was excluded because of its unstable remanence (Fig. 5). The remaining eight samples are composed of andesite (L–6, 7, and 8) and iron-formation (L–9, 11, 13, 14, and 16). The magnetic characteristics of the samples yield an extremely broad range of values (Table II), including both stable and less stable remanences (Fig. 5, samples L–4 and 11), which are suggestive of both c.r.m. and

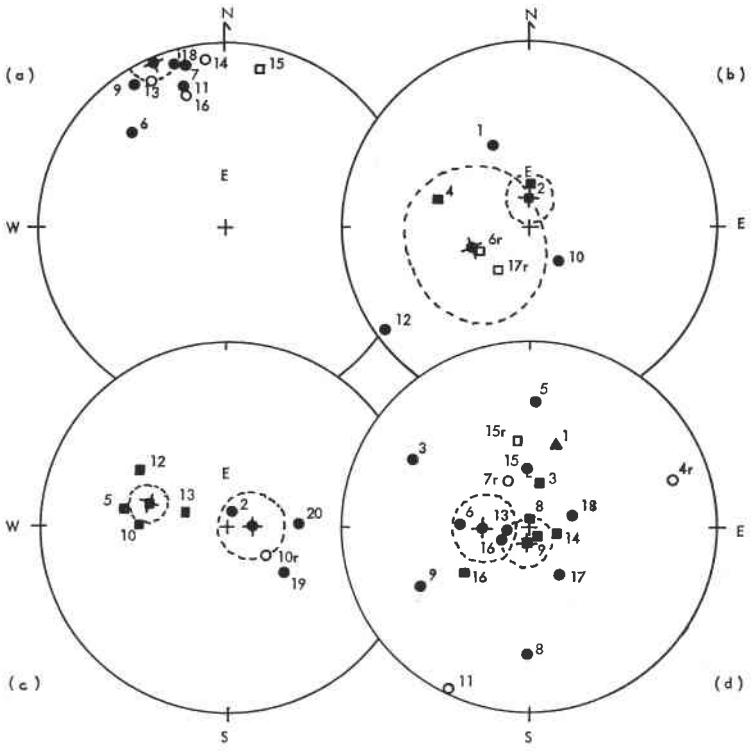


FIG. 4. Remanence directions for samples from: *a*, the Soudan iron-formation (○, ●) and Ogishke conglomerate (□) from surface after applying tectonic corrections; *b*, the Ely greenstone (●) from the Pioneer Mine and the andesites of the Soudan iron-formation (□, ■) from the Soudan Mine before applying tectonic corrections; *c*, the 'paint rock' or altered andesite (○ ●) from the Pioneer Mine and the 'altered andesite' (■) from the Soudan Mine before applying tectonic corrections; and *d*, the hard hematite ore from the Pioneer (○, ●) and Soudan (□, ■) Mines and the jasper (▲) from the Soudan Mine before applying tectonic corrections. The directions are plotted on Schmidt equal-area polar projections viewed from above with declinations reckoned clockwise from the north arrow and with positive (negative) inclinations directed downwards (upwards) and represented by solid (open) symbols. The symbol E indicates the present direction of the earth's magnetic dipole field. The mean direction for a group of remanence directions is represented by a cross through the group symbol and the cone of the circular standard deviation by a broken line. The letter 'r' after the sample number indicates that the remanence direction was reversed before the group mean direction was calculated.

t.r.m. remanences. Unfolding causes the precision parameter for the remanence directions to increase from 5.9 to 13.1, which suggests that the Soudan iron-formation has a pre-folding remanence (Fig. 4*a*).

Five samples of andesite from the Soudan iron-formation (N–2, 4, 6, 11, and 17) were collected in the Soudan Mine (Fig. 3) from within a few feet of the ore. One sample (N–11) has a remanence direction that is erratic with respect to the remaining four samples, and it was excluded. Two samples (N–6 and 17) have remanence directions with reversed polarity (Fig. 4*b*), which indicates stability. Unfolding causes the precision parameter for the remanence directions to decrease from 8.8 to 8.3, which indicates that the remanence may have been acquired either before or after folding, or that it has a mixed origin.

TABLE II

Summary of the magnetic measurements for samples from the surface (L), the Pioneer Mine (M) and the Soudan Mine (N) of the Vermilion Range, Minnesota (longitude 92.1° W.; latitude 47.9° N.)

Sample Group	N	X	Ms	Mn/Ms	Mean Remanence Vector					Pole Position			
					R	Dm	Im	α_{63}	k	Dp	Ip	dp	dm
L – Ely greenstone	6(U)	1.3–2.3	.006–.077	1–40	4.6	310	85	21	3.5				
L – Soudan iron-fm.	8(C)	1.3–2 120	.011–65	1–40	7.4	338	3	9	13.1	239	39	5	9
L – Ogishke cgl.	1(C)	.133	.019	6		15	−15						
M – Ely greenstone	3(U)	3.2–5.3	.014–.054	1–3	2.0	251	65	47	2.0				
M – 'Paint Rock'	4(U)	.95–15	1.1–14	1–1.5	3.7	90	79	15	9.8	63	44	27	28
M – Hard hematite ore	13(U)	18–173	2.6–16	1–9	9.2	257	71	15	3.2	132	32	23	27
N – Soudan IF–andesite	4(U)	2.4–5.0	.087–1.1	1–2	3.7	357	74	16	8.8				
N – Altered andesite	4(U)	4.2–4.5	.62–1.9	1–14	3.9	286	54	8	29.5	167	35	8	11
N – Soudan IF–jasper	1(U)		1.8	3		154	37						
N – Hard hematite ore	6(U)	4.9–24	2.6–60	1	5.6	182	83	11	11.2	93	34	21	21

NOTE: The symbols are N, the number of samples represented in the group or the population of the remanence directions; C(U), indicates that the remanence directions have been corrected (uncorrected) for tectonic deformation prior to calculation of the mean remanence direction and pole position; X, the range of susceptibilities in (e.m.u. × 10^6/g); Ms, the range of intensities of the stable remanences in (e.m.u. × 10^5/g) after a-c. demagnetization at 700 Oe (peak); Mn/Ms, the range of ratios of n.r.m. intensity to Ms intensity, which roughly indicates the ratios of unstable (Mn–Ms) to stable (Ms) remanence; R, the length of the mean remanence vector with each of the N remanence directions considered as a unit vector; Dm, the declination of the mean remanence vector in degrees reckoned clockwise from 0°; Im, the inclination of the mean remanence vector in degrees reckoned from the horizontal plane and considered positive (negative) if directed downwards (upwards); α_{63}, the semi-angle of the circular standard deviation, in degrees, of the cluster of remanence directions; k, the precision parameter of the cluster of remanence directions; $Dp(Ip)$, the longitude (latitude) of the virtual paleomagnetic pole position in degrees; $dp(dm)$, the semi-axis of the oval of confidence derived from the circular standard deviation perpendicular to (along) the meridian between the sampling site and the pole position in degrees about the pole position.

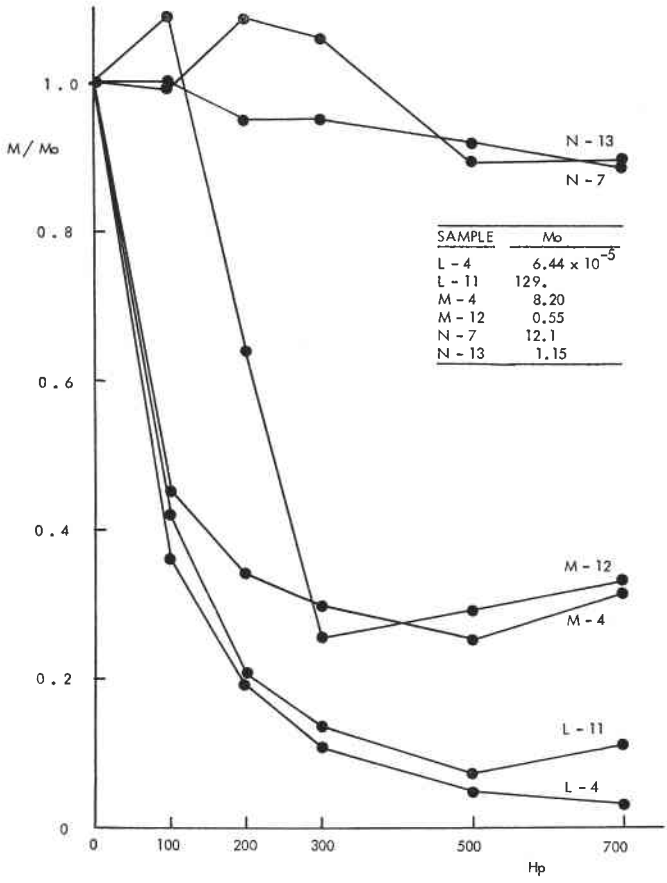


FIG. 5. A-c. demagnetizing curves for which M/Mo is the relative remanence intensity with Mo in (e.m.u./g) and Hp is the intensity of the alternating demagnetizing field in oersteds (peak). From polished sections, sample L–4 of andesite of the Soudan iron-formation contains a few disseminated grains of corroded pyrite; sample L–11 of Soudan iron-formation contains a few euhedral grains of magnetite disseminated through the gangue; sample M–4 of hard hematite ore contains hematite in irregularly shaped masses and randomly oriented laths with a few corroded inclusions of magnetite; sample M–12 of Ely greenstone contains a few euhedra of pyrite and a trace amount of hematite–limonite 'dust' in the gangue; sample N–7 of hard hematite ore contains massive hematite with a very few inclusions of gangue and magnetite, but without preferential orientation or banding; and, sample N–13 of 'altered andesite' contains a few coarse grains of martite with magnetite cores and a few corroded hematite grains.

The one sample (N–1) of jasper from the Soudan iron-formation adjacent to the ore in the Soudan Mine (Fig. 3) has a moderately stable remanence with a direction that is most similar to those of the hard hematite ore (Fig. 4*d*).

Altered Andesite Wall-Rocks

Immediately adjacent to the hard hematite ore of the Pioneer Mine (Fig. 2), the Ely greenstone has been thoroughly altered to 'paint rock' or kaolin impregnated with hematite. Four samples of 'paint rock' (M–2, 10, 19, and 20) were measured. These samples proved to have extremely stable remanences

(Table II) of low intensity, which probably indicate a c.r.m. Unfolding causes the precision parameter for the remanence directions to decrease from 9.8 to 1.4, which strongly suggests a post-folding remanence as shown in Fig. 4*c*. One sample (M–10) has a reversed remanence polarity, which aligns with the post-folding group to add further credence to the interpretation of a post-folding remanence for the altered andesites.

Immediately adjacent to the hard hematite ore of the Soudan Mine (Fig. 3), the andesites of the Soudan iron-formation have been leached, sericitized, and impregnated with minor amounts of hematite. The four samples of this altered andesite (N–5, 10, 12, and 13) have extremely stable remanence moments of low intensity (Fig. 5, sample N–13) and very uniform magnetic characteristics (Table II), which are probably indicative of a c.r.m. Unfolding causes the precision parameter for the remanence directions to decrease from 29.5 to 17.3, which is indicative of a post-folding remanence as shown in Fig. 4*c*.

Hard Hematite Ore

At Ely, the ore is composed of small angular fragments of hard blue hematite that are cemented together by secondary hematite with pockets of soft hematite and magnetite. The hematite crystals are randomly oriented. The ore is cut by minor altered intrusive rocks and by late joints with secondary minerals such as carbonate, quartz, pyrite, chalcopyrite, and apatite. The deposit exhibits mineral zoning with temperature eutectics between 200 and 350 °C. Fourteen samples (M–3 to 9, 11, and 13 to 18) of the hard hematite ore were collected (Fig. 2), of which one sample (M–14) crumbled and could not be used. The remaining 13 samples have very stable remanences of moderate intensity after a-c. demagnetizing (Table II; Fig. 5, sample M–4), which probably indicates a c.r.m. The moderately intense unstable remanence associated with the natural remanent magnetization (n.r.m.) is probably caused by minute magnetite inclusions in the hematite, which appear to be altering to hematite and which also suggests a c.r.m. Two samples (M–4 and 7) have a reversed remanence polarity prior to unfolding (Fig. 4*d*). Unfolding causes the precision parameter for the remanence directions to decrease from 3.2 to 1.2, which indicates a post-folding remanence.

At Soudan, the hard hematite ore is dense, blue, and very fine grained, with minor amounts of martite and magnetite. In places, the ore is faintly banded, which has been interpreted as a relic of the primary banding in the replaced iron-formation. Late quartz-filled veinlets cut the ore. Small erratic pockets of chalcopyrite, native copper, cuprite, pyrite, quartz, and specular hematite are found in the ore deposits. Mineral zoning has not been noted in the ore deposits. Seven samples of the hard hematite ore (N–3, 7, 8, 9, 14, 15, and 16) were collected (Fig. 3). They have extremely stable remanence moments (Fig. 5, sample N–7) of moderate intensity (Table II), which strongly suggests a c.r.m. One sample (N–7) gives a remanence direction that diverges by 70° of arc from the remaining six sample remanence directions and therefore it was excluded. One sample (N–15) has a remanence direction with reversed polarity in the post-folding remanence direction group (Fig. 4*d*). Unfolding causes the precision

parameter to decrease from 11.2 to 4.7, which strongly suggests a post-folding remanence.

Ogishke Conglomerate

The Knife Lake slates contain several thousand feet of slates and graywackes intercalated with numerous conglomerates and tuffs, and with minor agglomerates and flows. Generally, a basal conglomeratic member, the Ogishke conglomerate, is present. One graywacke sample (L–15) was collected on surface (Fig. 1) from the Ogishke conglomerate. It has a moderately stable remanence (Table II), which is most similar in direction (Fig. 4*a*) to the pre-folding remanence directions of the Soudan iron-formation and which probably results from a primary detrital remanent magnetization (d.r.m.).

INTERPRETATION AND CONCLUSIONS

The samples from the Soudan iron-formation on surface have a stable pre-folding remanence, which is probably a c.r.m. for the iron-formation and a t.r.m. for the andesite. Their remanence direction gives a pole position (Table II) that is similar to those reported by Symons (1967) for Early Precambrian iron-formation of the nearby Steep Rock Group, and by Strangway (1964) for Early Precambrian basic dikes in the Superior structural Province. The Ely greenstone on surface is also thought to have a stable pre-folding remanence, probably a t.r.m., because it is probably similar in age to the Soudan iron-formation, gives a similar post-folding remanence direction to the Soudan iron-formation, and has a low post-folding value for its precision parameter. The Ogishke conglomerate is also thought to have a stable pre-folding remanence, probably a d.r.m., which differs slightly in direction from that of the Soudan iron-formation because of the time interval between their deposition.

The samples of Ely greenstone and of andesites of the Soudan iron-formation that were collected from near the ore in the Pioneer and Soudan Mines respectively, are interpreted as having a mixed remanence. This remanence is composed of a primary t.r.m. related to their original deposition and a secondary t.r.m. or c.r.m. related to heat or chemical reaction during ore deposition. A mixed remanence would explain the inconclusiveness of the fold test.

The remanence of the altered andesites from adjacent to the ore (i.e. the 'paint rock' of the Pioneer Mine and the 'altered andesite' of the Soudan Mine) is post-deformational in origin, aligns in direction with the hard hematite ore direction, and is probably a c.r.m. Clearly, the remanence of the altered andesites was formed at the time of ore deposition.

In both deposits, the hard hematite ore has a stable post-folding remanence which is probably a c.r.m., as is clearly shown for the Soudan Mine. The low value of the precision parameter for the ore at the Pioneer Mine is attributed to minor brecciation of the ore after its formation. The ore's remanence directions and pole positions (Table II) are similar to those reported for the Lower to Middle Keweenawan (e.g. P. M. DuBois (1962) and Symons (1965)), which corresponds to a major igneous period in the geological history of the Vermilion Range. The jasper, or altered Soudan iron-formation adjacent to the ore, is

thought to have a post-folding remanence, which is similar in direction and genetically related to the ore's remanence.

In conclusion, these paleomagnetic results are consistent with the genetic hypothesis that: (*a*) the Ely greenstone, Soudan iron-formation, and Ogishke conglomerate were deposited with primary t.r.m., c.r.m., and d.r.m. respectively, and subsequently folded; and, (*b*) the hard hematite ore was epigenetically emplaced in the fold structures with a post-folding c.r.m., which is duplicated in the heavily altered adjacent wall-rocks of andesite (i.e. 'paint rock' and 'altered andesite') and iron-formation (jasper). During ore emplacement, a secondary remanence component, which might be a t.r.m. resulting from heat or a c.r.m. resulting from the chemical activity of solutions, or both, was superimposed on the primary remanence of the andesite wall-rocks within a few feet of the ore. Thus the genetic hypotheses of (1) Clements (1903) and Abbott (1907), (2) Gruner (1926, 1937), and (3) Roberts and Bartley (1943), Schwartz and Reid (1955), Reid (1956*a*, 1956*b*), Klinger (1956) and Machamer (1964), are all tenable. The geologic evidence from studies on the ore's mineralogy, texture, and structure seems to favour alternatives (2) and (3), which consider that the primary magnetite-banded and hematite-banded oxide facies of the Soudan iron-formation were: (2) leached and oxidized by hydrothermal solutions to leave hard hematite; and, (3) replaced by ascending iron-bearing hydrothermal solutions to form hard hematite.

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